

Vehicle-Resale Data – Question Bank

Below are **20 questions** designed to test pandas skills (EDA, filtering, grouping, aggregation, and time-series operations).

Each question is presented in a markdown table with the **required output description**.

1 Exploratory Data Analysis (EDA)

#	Question	Expected Output
Q1	How many records are there per model Year ?	Table (or Series) indexed by Year with a column Count showing the number of rows for each year, sorted chronologically.
Q2	What is the distribution of vehicle age at sale (Age = Sale Year – Model Year)? Provide basic statistics (count, mean, median, std, min, 25 %, 50 %, 75 %, max).	Summary table (similar to <code>describe()</code>) for the derived column Vehicle Age containing the listed statistics.
Q3	Which Body styles are most common? List the top-5 body styles with their frequencies.	Table with two columns: Body and Frequency , showing the five most frequent body styles ordered from highest to lowest.
Q4	What are the minimum , maximum , and average odometer readings across the whole dataset?	Single-row summary (or three-row table) with columns Min Odometer , Max Odometer , and Mean Odometer .
Q5	How many unique sellers exist in each State ?	Table indexed by State with a column Seller Count indicating the number of distinct Seller values per state.

2 Filtering

#	Question	Expected Output
Q6	Show all rows where Selling Price > Market Price .	Filtered dataset (all original columns) containing only records that satisfy the condition.
Q7	Retrieve automatic-transmission vehicles in California with Odometer < 30 000 miles.	Subset of the original data (all columns) limited to the specified criteria.
Q8	For each Make , keep the record with the most recent Sale Date .	Table with one row per Make , preserving all columns of the latest-dated sale for that make.

#	Question	Expected Output
Q9	Return vehicles sold in the last 90 days of the entire dataset timeline.	Subset (all columns) where Sale Date $\geq$ (maximum Sale Date – 90 days).
Q10	List transactions where Selling Price exceeds the 90th percentile of all selling prices.	Filtered dataset representing the top 10% most expensive sales, with all original columns retained.

3 Group-by & Aggregation

#	Question	Expected Output
Q11	Compute the average Selling Price and average Odometer for each State .	Table indexed by State with columns Avg Selling Price and Avg Odometer .
Q12	For each Make , calculate the total units sold (count of VIN) and the total revenue (sum of Selling Price).	Table indexed by Make with columns Count and Revenue .
Q13	Compute Delta = Selling Price – Market Price and find the median Delta for each Body style.	Table indexed by Body with a column Median Delta .
Q14	Produce a quarter-wise (3-month) count of sales transactions across the whole dataset.	Time-series table where each row represents a calendar quarter (e.g., 2020-Q1) with column Sales Count .
Q15	For each State , calculate annual revenue (sum of Selling Price) and the year-over-year growth % .	Multi-level table indexed by State and Year with columns Annual Revenue and YoY Growth % .

4 Timeseries Operations (Resampling)

#	Question	Expected Output
Q16	Resample the data to a monthly frequency and compute the mean Selling Price for each month.	Time-series table indexed by month (e.g., 2020-01) with column Avg Monthly Price .
Q17	Resample to a weekly frequency (weeks ending on Sunday) and count the number of sales per week.	Time-series table indexed by week-ending dates with column Weekly Sales .
Q18	Calculate a centered 6-month rolling average of Odometer based on Sale Date .	Time-series table with Sale Date (or month) as index and column 6-Month Rolling Odo Avg .

#	Question	Expected Output
Q19	For each month, extract the Selling Price of the first sale (minimum Sale Date within that month).	Table indexed by month with column First-Sale Price (single value per month).
Q20	Generate a cumulative revenue series ordered by Sale Date .	Time-ordered table with Sale Date and column Cumulative Revenue that monotonically increases.

All questions are written to be answered with pandas operations only; the resulting tables can then be visualised with Plotly Express
